

Quantified Statements

Sehyun Kwon

Mathematical Data Science
Hanyang University ERICA

Review

Let $P(x)$ be a predicate on a universe U .

The proposition $(\forall x \in U)P(x)$ is true exactly when $P(a)$ is true for every $a \in U$.

The proposition $(\exists x \in U)P(x)$ is true exactly when $P(a)$ is true for at least one $a \in U$.

A counterexample refutes a universal proposition; a witness establishes an existential proposition.

The prime-testing problem

Recall the predicates on $\mathbb{Z}$:

$$C(n) = \text{"check_prime}(n) \text{ returns } \mathbf{T}"$$

and

$$P(n) = \text{"}n \text{ is prime."}$$

The correctness claim is

$$(\forall n \in \mathbb{Z})(C(n) \Leftrightarrow P(n)).$$

Its justification involves both an input n and a possible divisor of n .

Scope

The **scope** of a quantifier is the part of a statement controlled by that quantifier.

In

$$(\forall x \in \mathbb{Z}) \underbrace{(x < 0 \Rightarrow x^2 > 0)}_{\text{scope of } \forall x},$$

the scope of $\forall x$ is the expression inside the parentheses.

The variable x is bound throughout that scope.

Parentheses make the scope explicit.

Nested quantifiers

Let $R(x, y)$ be a predicate for $x, y \in \mathbb{Z}$.

The proposition

$$(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})R(x, y)$$

says:

For every integer x , there is an integer y for which $R(x, y)$ is true.

The inner existential proposition is evaluated separately for each x .
Both variables are bound, so the entire statement is a proposition.

Quantifier order

Define

$$R(x, y) = "x < y", \quad x, y \in \mathbb{Z}.$$

The proposition

$$(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})R(x, y)$$

is true: after x is given, choose $y = x + 1$.

The proposition

$$(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})R(x, y)$$

is false: no single integer is larger than every integer.

Choosing the existential value

Let $R(x, y)$ be a predicate on a universe U .

$$(\forall x \in U)(\exists y \in U)R(x, y)$$

For every x in U , there exists a y in U such that $R(x, y)$ is true.
Given any x , we can find a suitable y . The choice of y may depend on x .

$$(\exists y \in U)(\forall x \in U)R(x, y)$$

There exists a y in U such that, for every x in U , $R(x, y)$ is true.
A single choice of y must work for all x .

Equality over the integers

Let $E(x, y)$ be the predicate “ $x = y$,” where $x, y \in \mathbb{Z}$. In this slide, omitted universes are $\mathbb{Z}$.

$$(\forall x)(\forall y)E(x, y) \quad \mathbf{F}$$

$$(\forall x)(\exists y)E(x, y) \quad \mathbf{T}$$

$$(\exists x)(\forall y)E(x, y) \quad \mathbf{F}$$

$$(\exists x)(\exists y)E(x, y) \quad \mathbf{T}$$

For the second proposition, choose $y = x$. For the fourth, choose $x = y = 0$.

Repeated quantifiers

Let $R(x, y)$ be a predicate on a universe U .

Two universal quantifiers may be exchanged:

$$(\forall x \in U)(\forall y \in U)R(x, y) \equiv (\forall y \in U)(\forall x \in U)R(x, y).$$

Two existential quantifiers may also be exchanged:

$$(\exists x \in U)(\exists y \in U)R(x, y) \equiv (\exists y \in U)(\exists x \in U)R(x, y).$$

These equivalences do not permit exchanging $\forall$ with $\exists$.

Review: the prime-testing algorithm

For an integer input n , `check_prime(n)` should return **T** exactly when n is prime.

```
def check_prime(n):  
    if n <= 1:  
        return False  
    i = 2  
    while i * i <= n:  
        if i divides n:  
            return False  
        i = i + 1  
    return True
```

Here “ i divides n ” means that n is an integer multiple of i ; the code is pseudocode.

The loop tests $i = 2, 3, \dots$ only while $i^2 \leq n$.

Why can we stop there without missing a composite number?

Square roots

For a positive real number n , the notation $\sqrt{n}$ denotes the nonnegative real number whose square is n .

Recall the loop condition `while i * i <= n`. For $n > 1$ and $i \geq 2$,

$$i^2 \leq n \iff i \leq \sqrt{n}.$$

Thus the loop checks possible divisors only up to $\sqrt{n}$.

Claim to prove: Every composite integer n has an integer divisor d satisfying $2 \leq d \leq \sqrt{n}$.

This would ensure that every composite input is detected before the loop ends.

The companion factor

Let n be a positive composite integer, and let a be a positive divisor satisfying

$$\sqrt{n} < a < n.$$

Since $a \mid n$, define the integer

$$b = \frac{n}{a}.$$

Because $a < n$, we have $b > 1$, hence $b \geq 2$.

Because $a > \sqrt{n}$,

$$b = \frac{n}{a} < \frac{n}{\sqrt{n}} = \sqrt{n}.$$

The quantified factor statement

Define

$$\mathbb{Z}^+ = \{1, 2, 3, \dots\}.$$

For $n, a \in \mathbb{Z}^+$, define

$$K(n) = "n \text{ is composite}", \quad D(a, n) = "a \mid n."$$

The preceding argument establishes

$$\begin{aligned} (\forall n \in \mathbb{Z}^+)(\forall a \in \mathbb{Z}^+)(K(n) \wedge D(a, n) \wedge \sqrt{n} < a < n \\ \Rightarrow 2 \leq n/a < \sqrt{n}). \end{aligned}$$

Why a divisor is found

Let n be composite. Then n has a positive divisor a with $1 < a < n$.

If $a \leq \sqrt{n}$, the algorithm tests a .

If $a > \sqrt{n}$, its companion factor $b = n/a$ satisfies

$$2 \leq b < \sqrt{n},$$

so the algorithm tests b .

Therefore every composite input has a divisor among the integers tested.

Correctness of the prime test

If $n \leq 1$, the algorithm returns **F**, and n is not prime.

If n is prime, none of the tested integers divides n , so the algorithm returns **T**.

If n is composite, some tested integer divides n , so the algorithm returns **F**.

Hence

$$(\forall n \in \mathbb{Z})(C(n) \Leftrightarrow P(n))$$

is true.

Negating a universal proposition

Let the universe be $\mathbb{Z}^+$, and define

$$P(n) = "n \text{ is prime}."$$

The proposition $(\forall n \in \mathbb{Z}^+)P(n)$ is false because $P(4)$ is false.
Its negation says that some positive integer is not prime:

$$(\exists n \in \mathbb{Z}^+)\neg P(n).$$

In general,

$$\neg(\forall x \in U)P(x) \equiv (\exists x \in U)\neg P(x).$$

Negating an existential proposition

Let the universe be $\mathbb{Z}$, and define

$$S(x) = "x^2 < 0."$$

The proposition $(\exists x \in \mathbb{Z})S(x)$ is false.

Its negation says that every integer has a nonnegative square:

$$(\forall x \in \mathbb{Z})\neg S(x).$$

In general,

$$\neg(\exists x \in U)P(x) \equiv (\forall x \in U)\neg P(x).$$

Negating a universal implication

Let $P(x)$ and $Q(x)$ be predicates on U .

Move the negation across the universal quantifier:

$$\neg(\forall x \in U)(P(x) \Rightarrow Q(x)) \equiv (\exists x \in U)\neg(P(x) \Rightarrow Q(x)).$$

An implication is false exactly when its hypothesis is true and its conclusion is false. Therefore,

$$\neg(\forall x \in U)(P(x) \Rightarrow Q(x)) \equiv (\exists x \in U)(P(x) \wedge \neg Q(x)).$$

Negating nested quantifiers

Let $R(x, y)$ be a predicate on U .

Negate one quantifier at a time:

$$\begin{aligned}\neg(\forall x \in U)(\exists y \in U)R(x, y) \\ &\equiv (\exists x \in U)\neg(\exists y \in U)R(x, y) \\ &\equiv (\exists x \in U)(\forall y \in U)\neg R(x, y).\end{aligned}$$

Thus every quantifier changes type when the negation moves through it.

Negating correctness

The correctness claim is

$$(\forall n \in \mathbb{Z})(C(n) \Leftrightarrow P(n)).$$

Its negation is

$$(\exists n \in \mathbb{Z}) \neg (C(n) \Leftrightarrow P(n)).$$

In words, the algorithm is incorrect exactly when there is an integer on which its answer disagrees with primality.

Such an input is a counterexample to correctness.

Summary

- ▶ The scope of a quantifier determines the variables it binds.
- ▶ The order of $\forall$ and $\exists$ generally changes the meaning.
- ▶ Every composite integer has a factor no greater than its square root.
- ▶ Negation changes $\forall$ to $\exists$, and $\exists$ to $\forall$.
- ▶ A counterexample is a witness for the negation of a universal claim.